Checkpoint 1

Total points 5/6

Dikerakan yagesya biar makin paham. Santai aja gausah dipikir berat.

Nama * ···/1

Brilliano Putra Pradhitya

Which are the two common types of supervised learning? (Choose two) *

- [ ] Clustering
- [x] Regression
- [x] Classification

Which of these is a type of unsupervised learning? * 1/1

- Clustering
- Classification
- Regression

For linear regression, the model is $f_{w,b}(x) = wx + b$. Which of the following are the inputs, or features, that are fed into the model and with which the model is expected to make a prediction? * 1/1

For linear regression, the model is $f_{w,b}(x) = wx + b.$

- x
- w and b
- (x, y)
- m

For linear regression, if you find parameters w and b so that J(w,b) is very close to zero, what can you conclude? *1/1

- ○ The selected values of the parameters w and b cause the algorithm to fit the training set really poorly.
- ◉ The selected values of the parameters w and b cause the algorithm to fit the training set really well. ✓
- ○ This is never possible — there must be a bug in the code.

* 1/1

Gradient descent is an algorithm for finding values of parameters w and b that minimize the cost function J.

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b)$$

When $\frac{\partial J(w,b)}{\partial w}$ is a negative number (less than zero), what happens to w after one update step?

- ( ) It is not possible to tell if w will increase or decrease
- (•) Increases ✓
- ( ) Stays the same
- ( ) Decreases

For linear regression, what is the update step for parameter b? *

1/1

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

A

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)})$$

B